

AviLight Ecological Impact and Diagnostics Report

Executive Summary

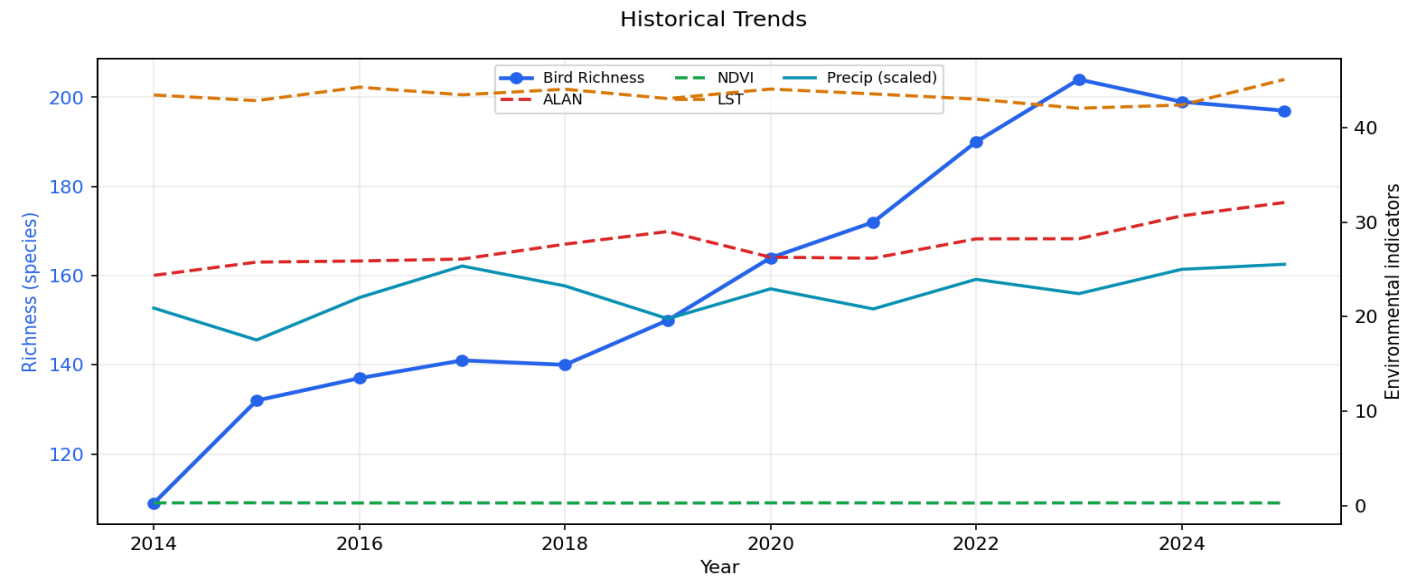
Run Context

Field	Value
Area	All Areas
Trend Years	2014-2025
Snapshot	2024-01
Generated	2026-04-18 20:45:26 UTC

AviLight Ecological Impact and Diagnostics Report

Ecological Impact View

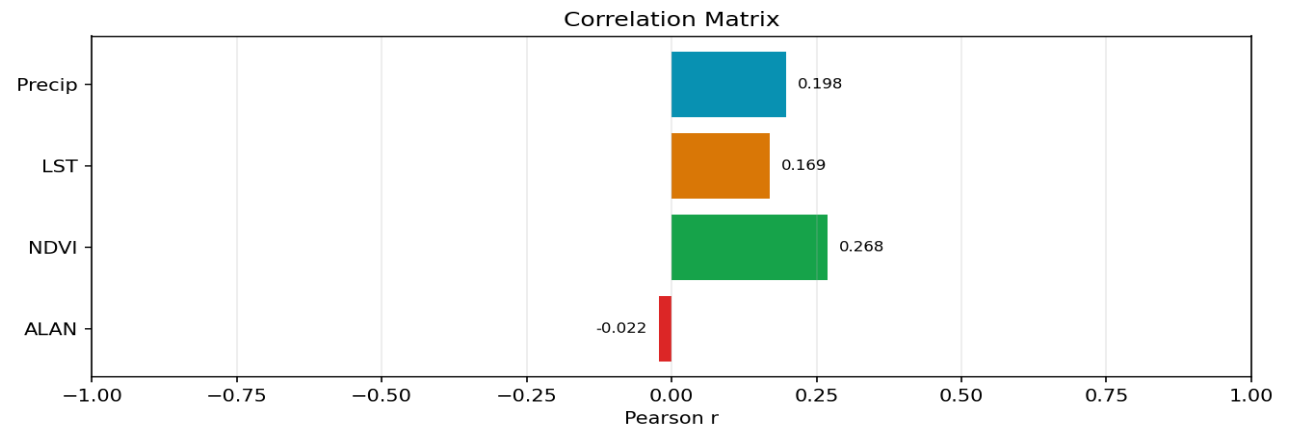
Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

Year	Richness	ALAN	NDVI	LST	Precip
2014	109.00	24.36	0.2843	43.45	2089.67
2015	132.00	25.77	0.2935	42.86	1752.42
2016	137.00	25.88	0.2766	44.28	2201.25
2017	141.00	26.09	0.2854	43.47	2535.23
2018	140.00	27.66	0.2721	44.06	2324.73
2019	150.00	29.00	0.2682	43.07	1977.59
2020	164.00	26.29	0.2856	44.08	2293.80
2021	172.00	26.18	0.2857	43.56	2081.15
2022	190.00	28.23	0.2724	43.02	2394.15
2023	204.00	28.24	0.2864	42.05	2242.81
2024	199.00	30.66	0.2810	42.39	2500.04
2025	197.00	32.07	0.2796	45.10	2553.47

Chart: Correlation Matrix

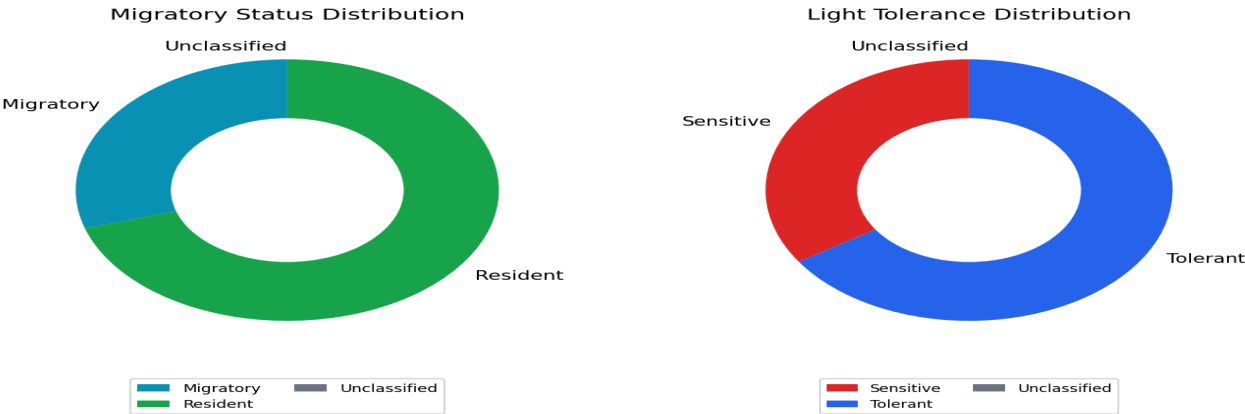


AviLight Ecological Impact and Diagnostics Report

Numerical Values: Correlation Coefficients

Metric	Pearson r
ALAN vs Richness	-0.0218
NDVI vs Richness	0.2683
LST vs Richness	0.1694
Precipitation vs Richness	0.1977

Chart: Migratory and Light Tolerance Distribution



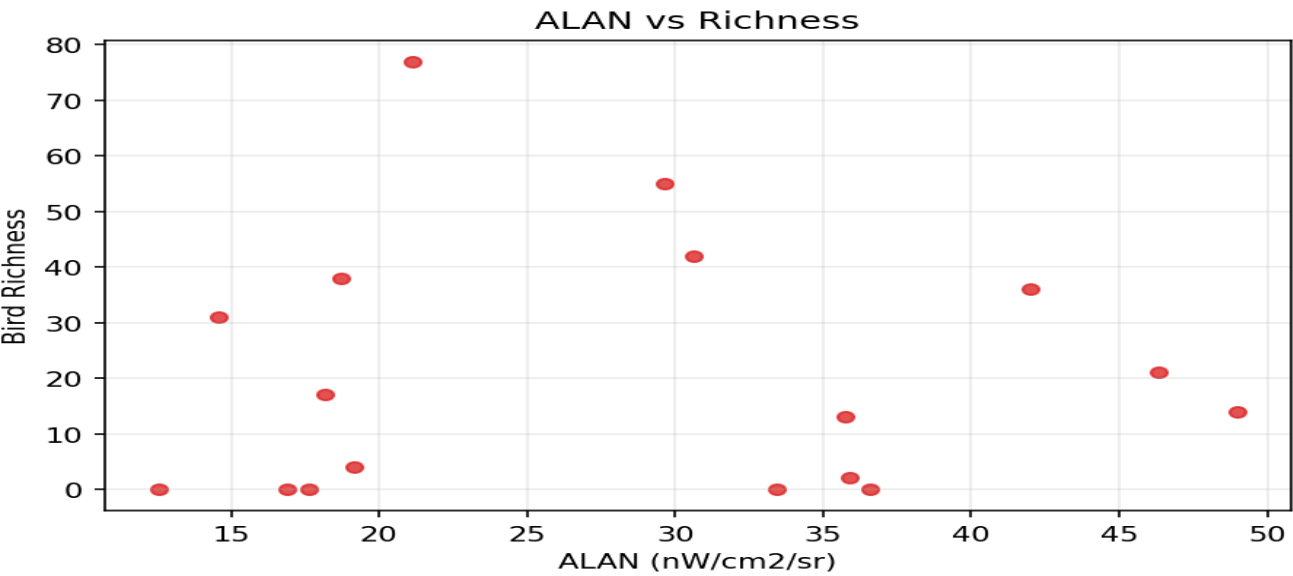
Numerical Values: Migration Distribution

Category	Count
Migratory	33
Resident	78
Unclassified	0

Numerical Values: Light Tolerance Distribution

Category	Count
Sensitive	38
Tolerant	73
Unclassified	0

Chart: ALAN vs Richness

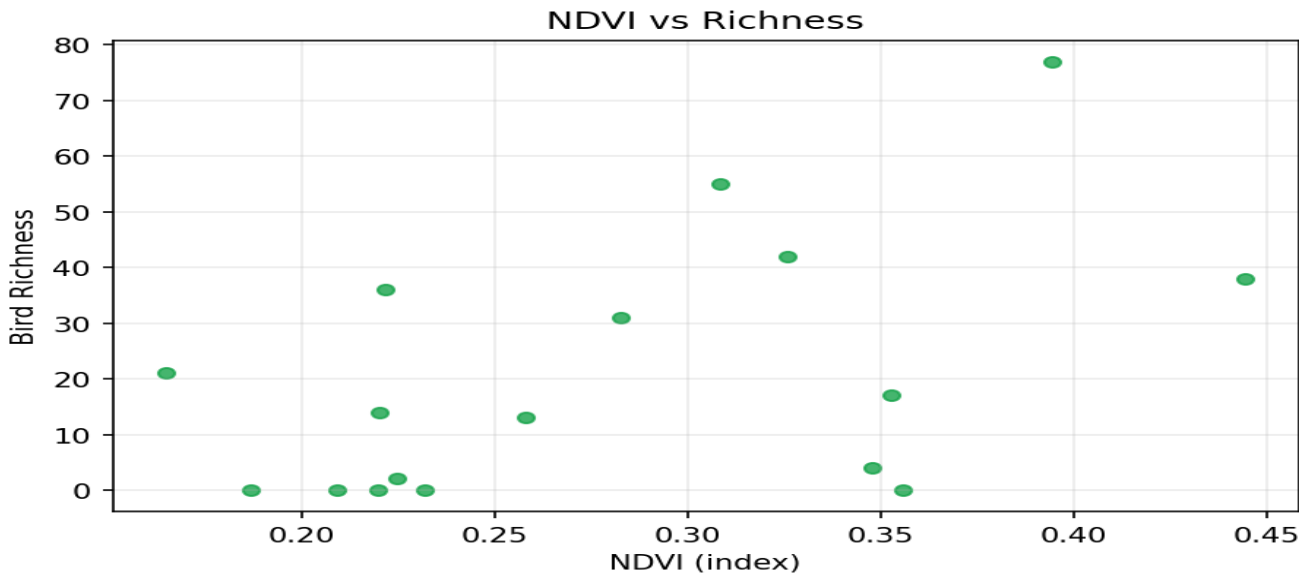


AviLight Ecological Impact and Diagnostics Report

Numerical Values: ALAN vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	17.6253	0.00
Las Piñas	Las Piñas	18.1804	17.00
Makati	Makati	42.0139	36.00
Malabon	Malabon	16.8709	0.00
Mandaluyong	Mandaluyong	35.8872	2.00
Manila	Manila	46.3457	21.00
Marikina	Marikina	19.1350	4.00
Muntinlupa	Muntinlupa	18.7014	38.00
Navotas	Navotas	12.5424	0.00
Parañaque	Parañaque	29.6273	55.00
Pasay	Pasay	48.9901	14.00
Pasig	Pasig	35.7328	13.00
Pateros	Pateros	33.4426	0.00
Quezon City	Quezon City	21.1494	77.00
San Juan	San Juan	36.5872	0.00
Taguig	Taguig	30.6320	42.00
Valenzuela	Valenzuela	14.5767	31.00

Chart: NDVI vs Richness



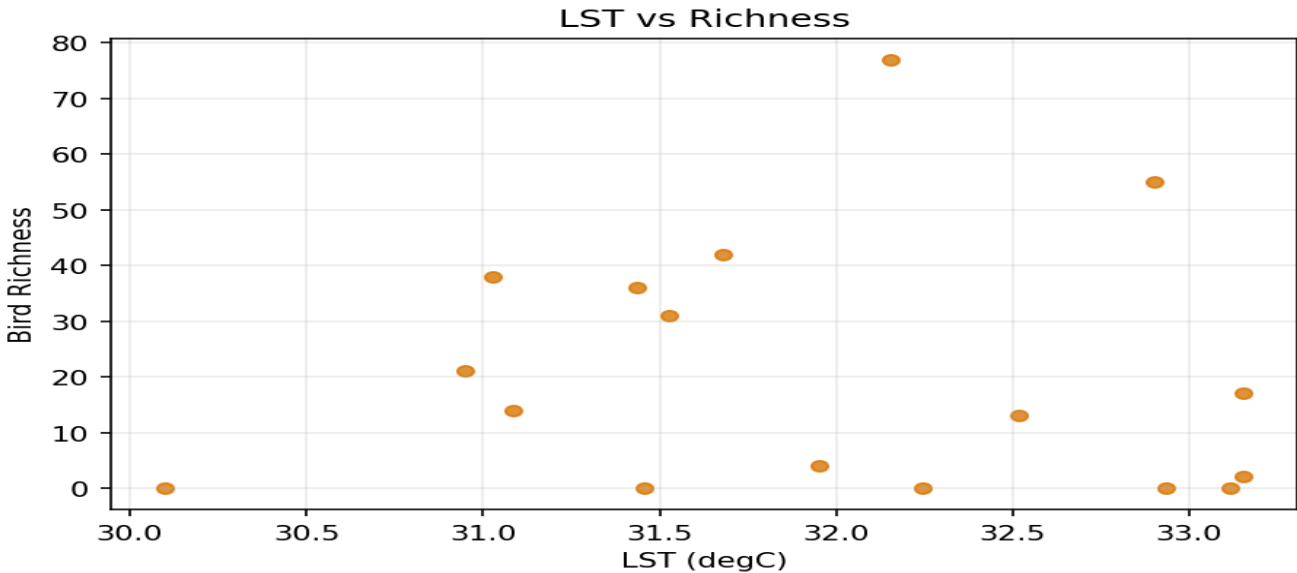
Numerical Values: NDVI vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	0.3558	0.00
Las Piñas	Las Piñas	0.3528	17.00
Makati	Makati	0.2215	36.00
Malabon	Malabon	0.2092	0.00
Mandaluyong	Mandaluyong	0.2247	2.00
Manila	Manila	0.1650	21.00
Marikina	Marikina	0.3479	4.00
Muntinlupa	Muntinlupa	0.4443	38.00
Navotas	Navotas	0.1869	0.00

AviLight Ecological Impact and Diagnostics Report

Parañaque	Parañaque	0.3084	55.00
Pasay	Pasay	0.2203	14.00
Pasig	Pasig	0.2579	13.00
Pateros	Pateros	0.2320	0.00
Quezon City	Quezon City	0.3943	77.00
San Juan	San Juan	0.2197	0.00
Taguig	Taguig	0.3258	42.00
Valenzuela	Valenzuela	0.2828	31.00

Chart: LST vs Richness

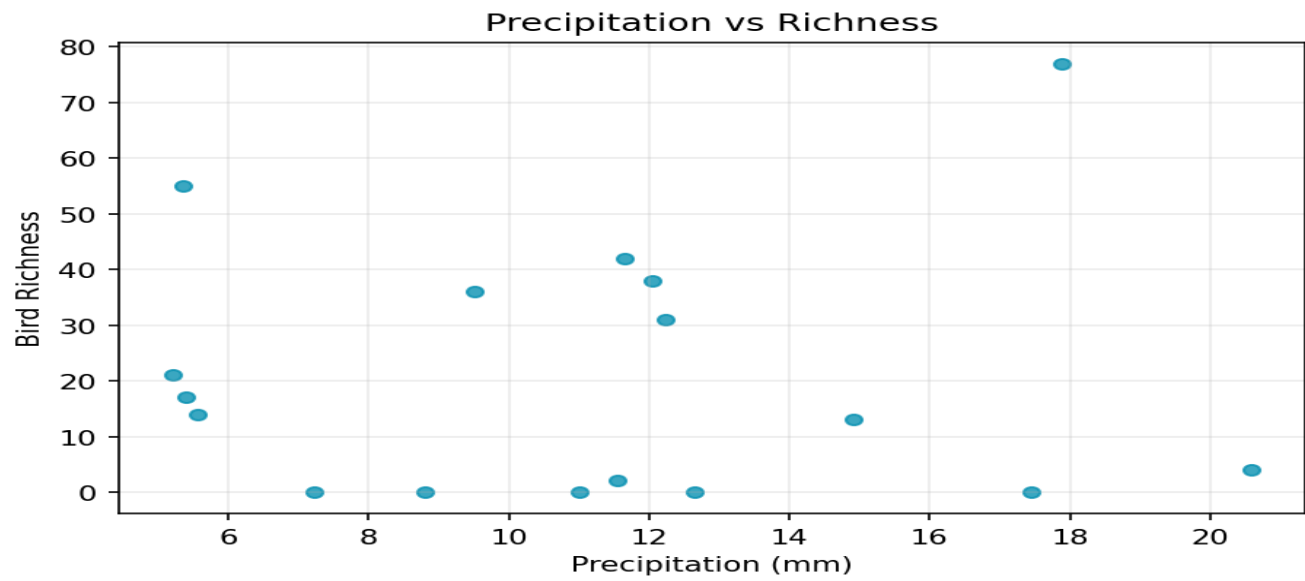


Numerical Values: LST vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	31.4546	0.00
Las Piñas	Las Piñas	33.1531	17.00
Makati	Makati	31.4346	36.00
Malabon	Malabon	30.0974	0.00
Mandaluyong	Mandaluyong	33.1521	2.00
Manila	Manila	30.9494	21.00
Marikina	Marikina	31.9531	4.00
Muntinlupa	Muntinlupa	31.0285	38.00
Navotas	Navotas	32.9352	0.00
Parañaque	Parañaque	32.9000	55.00
Pasay	Pasay	31.0866	14.00
Pasig	Pasig	32.5155	13.00
Pateros	Pateros	32.2433	0.00
Quezon City	Quezon City	32.1529	77.00
San Juan	San Juan	33.1166	0.00
Taguig	Taguig	31.6808	42.00
Valenzuela	Valenzuela	31.5286	31.00

AviLight Ecological Impact and Diagnostics Report

Chart: Precipitation vs Richness

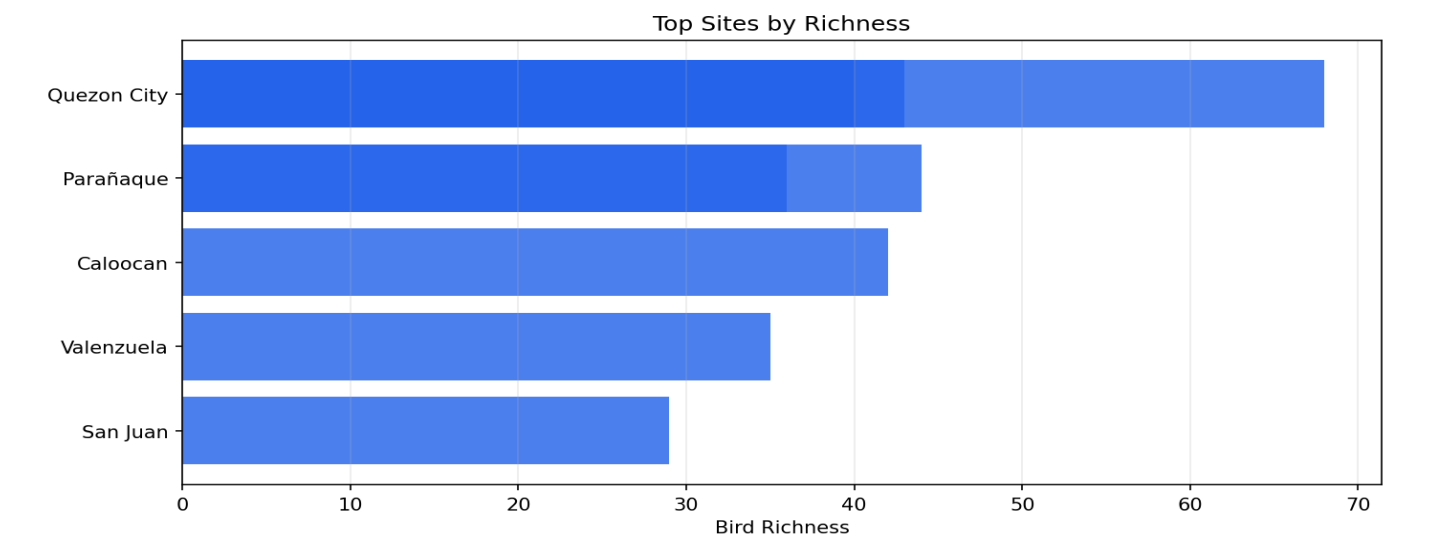


Numerical Values: Precipitation vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	17.4396	0.00
Las Piñas	Las Piñas	5.3914	17.00
Makati	Makati	9.5073	36.00
Malabon	Malabon	8.8099	0.00
Mandaluyong	Mandaluyong	11.5377	2.00
Manila	Manila	5.1975	21.00
Marikina	Marikina	20.5884	4.00
Muntinlupa	Muntinlupa	12.0530	38.00
Navotas	Navotas	7.2133	0.00
Parañaque	Parañaque	5.3558	55.00
Pasay	Pasay	5.5558	14.00
Pasig	Pasig	14.9217	13.00
Pateros	Pateros	12.6396	0.00
Quezon City	Quezon City	17.8904	77.00
San Juan	San Juan	11.0117	0.00
Taguig	Taguig	11.6590	42.00
Valenzuela	Valenzuela	12.2234	31.00

AviLight Ecological Impact and Diagnostics Report

Chart: Top 10 Sites by Richness



Numerical Values: Top Sites by Richness

Rank	Area	Richness
1	Quezon City	68
2	Parañaque	44
3	Quezon City	43
4	Caloocan	42
5	Quezon City	40
6	Quezon City	38
7	Parañaque	36
8	Valenzuela	35
9	Quezon City	34
10	San Juan	29

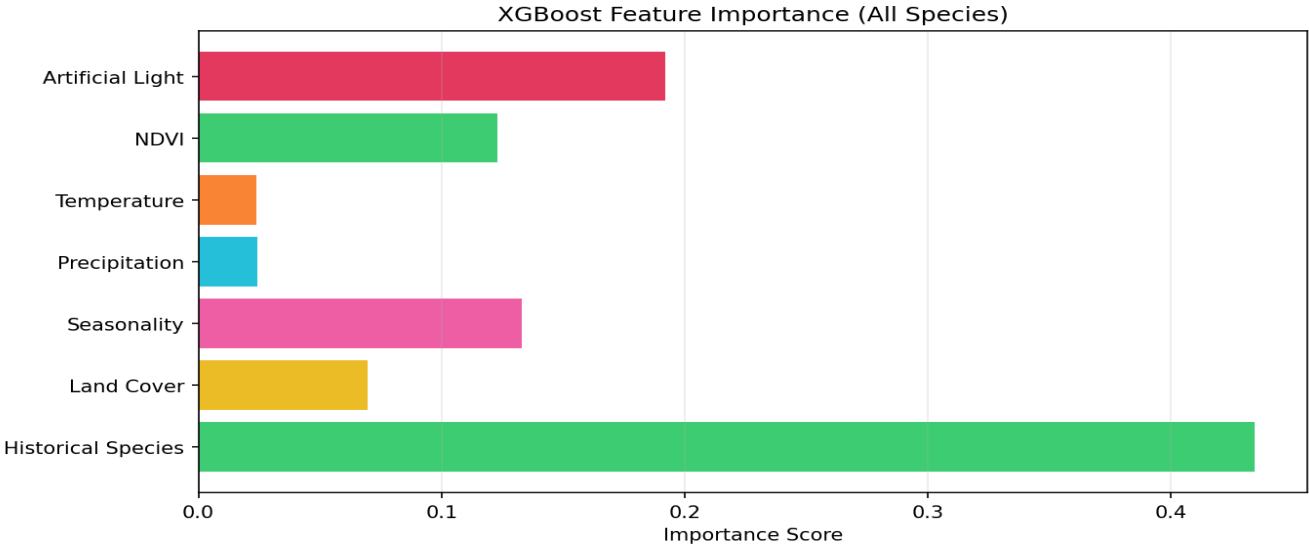
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Model Diagnostics View

KPI Cards

Metric	Value
RMSE	6.7345
MAE	4.1445
R2	0.7061

Chart: XGBoost Feature Importance (All Species)

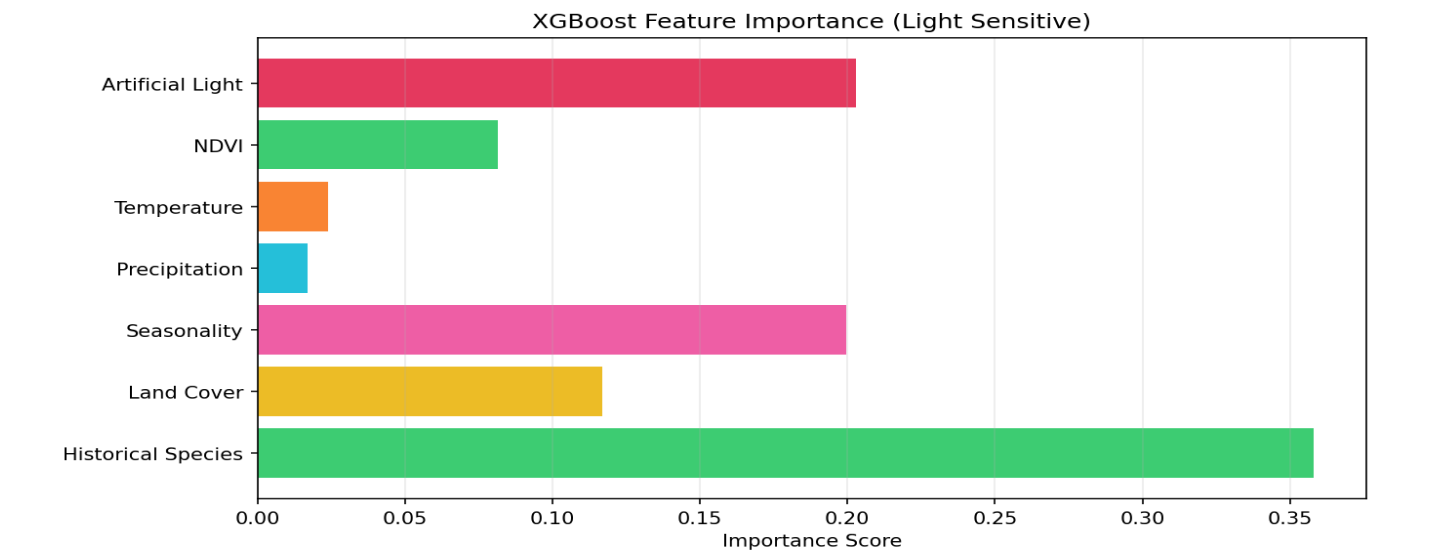


Numerical Values: XGBoost Feature Importance (All Species)

Feature	Importance
Artificial Light	0.191991
NDVI	0.122968
Temperature	0.023987
Precipitation	0.024173
Seasonality	0.133136
Land Cover	0.069444
Historical Species	0.434300

AviLight Ecological Impact and Diagnostics Report

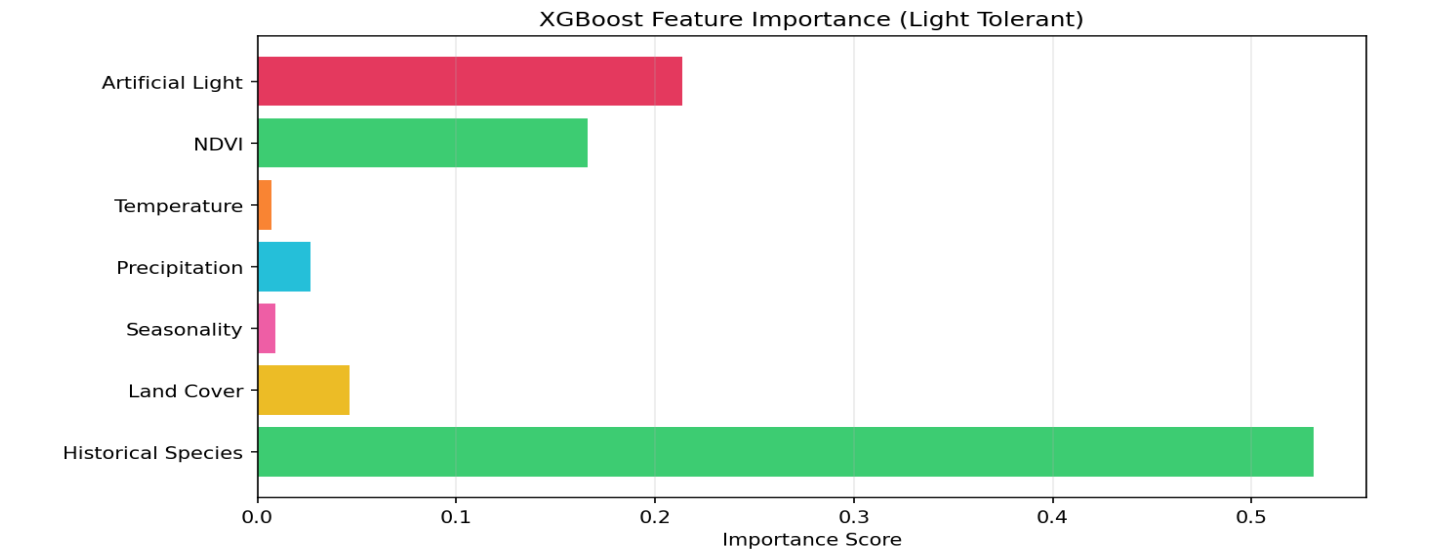
Chart: XGBoost Feature Importance (Light Sensitive)



Numerical Values: XGBoost Feature Importance (Light Sensitive)

Feature	Importance
Artificial Light	0.202727
NDVI	0.081639
Temperature	0.024060
Precipitation	0.017154
Seasonality	0.199484
Land Cover	0.117012
Historical Species	0.357924

Chart: XGBoost Feature Importance (Light Tolerant)



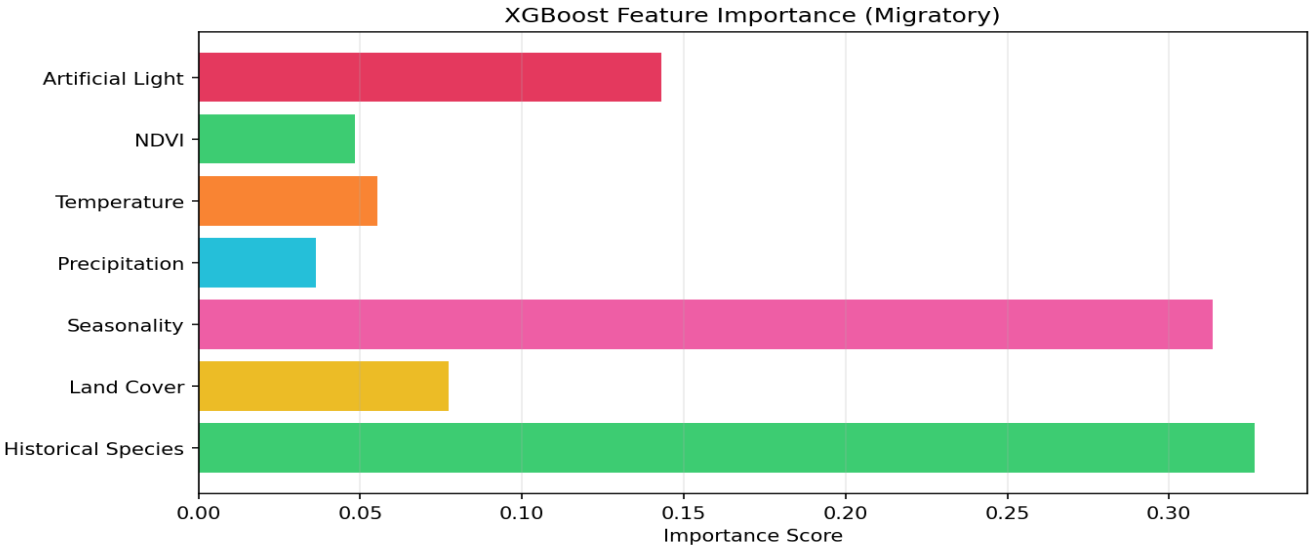
Numerical Values: XGBoost Feature Importance (Light Tolerant)

Feature	Importance
Artificial Light	0.213590
NDVI	0.166192
Temperature	0.007116

AviLight Ecological Impact and Diagnostics Report

Precipitation	0.026545
Seasonality	0.009277
Land Cover	0.046179
Historical Species	0.531101

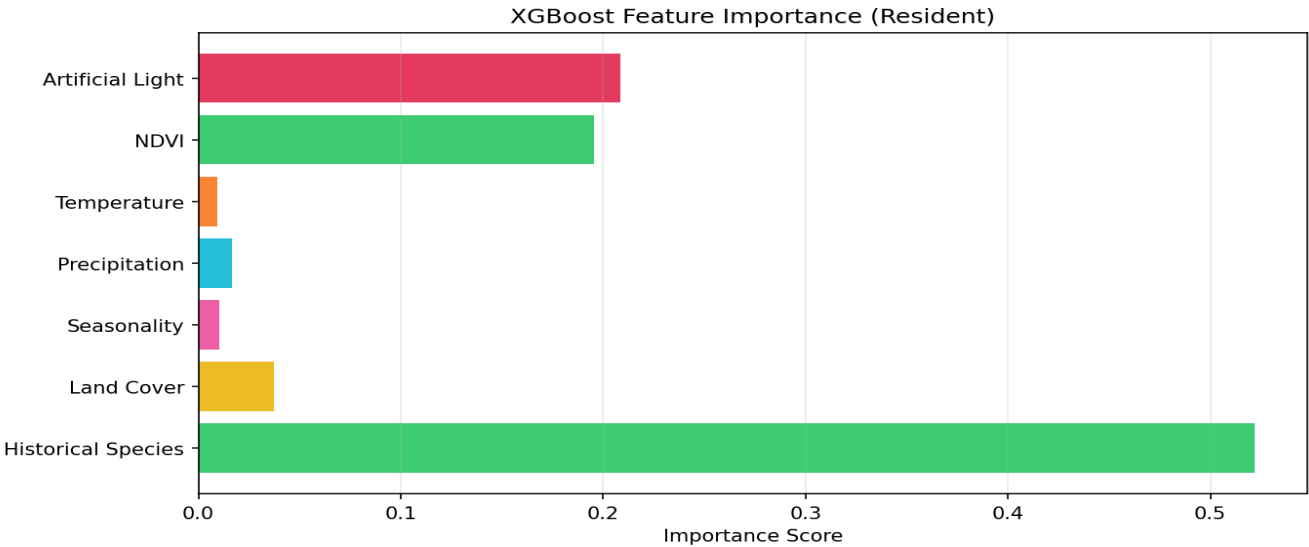
Chart: XGBoost Feature Importance (Migratory)



Numerical Values: XGBoost Feature Importance (Migratory)

Feature	Importance
Artificial Light	0.143060
NDVI	0.048403
Temperature	0.055268
Precipitation	0.036267
Seasonality	0.313430
Land Cover	0.077193
Historical Species	0.326379

Chart: XGBoost Feature Importance (Resident)

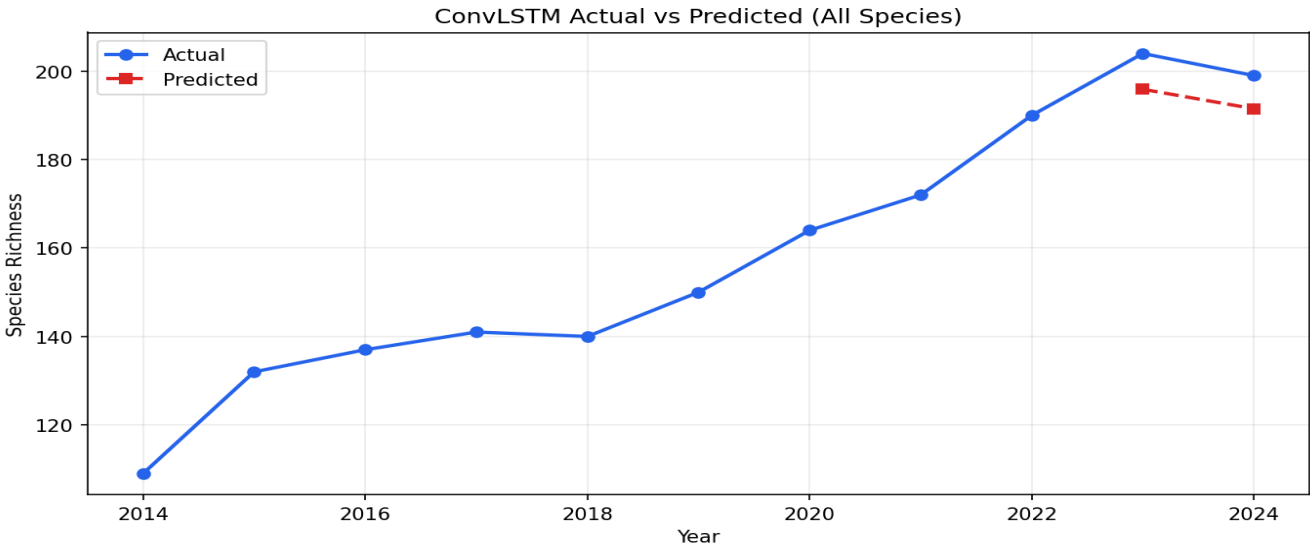


AviLight Ecological Impact and Diagnostics Report

Numerical Values: XGBoost Feature Importance (Resident)

Feature	Importance
Artificial Light	0.208588
NDVI	0.195638
Temperature	0.009504
Precipitation	0.016725
Seasonality	0.010355
Land Cover	0.037393
Historical Species	0.521797

Chart: ConvLSTM Actual vs Predicted (All Species)

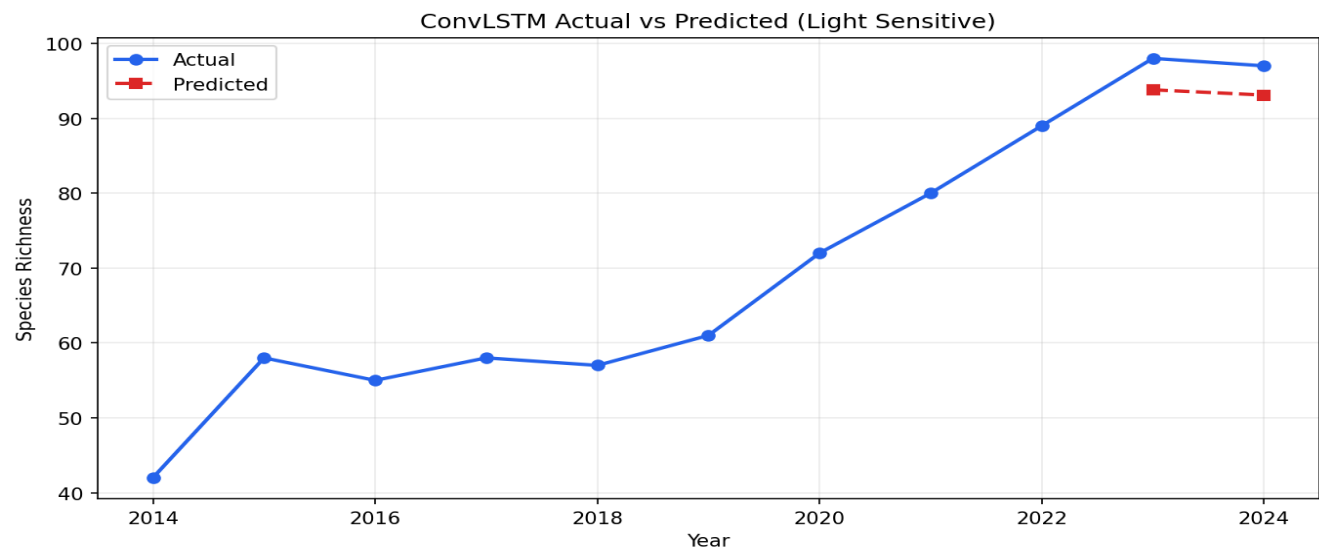


Numerical Values: ConvLSTM Actual vs Predicted (All Species)

Year	Actual	Predicted
2014	109.0000	-
2015	132.0000	-
2016	137.0000	-
2017	141.0000	-
2018	140.0000	-
2019	150.0000	-
2020	164.0000	-
2021	172.0000	-
2022	190.0000	-
2023	204.0000	195.9000
2024	199.0000	191.5000

AviLight Ecological Impact and Diagnostics Report

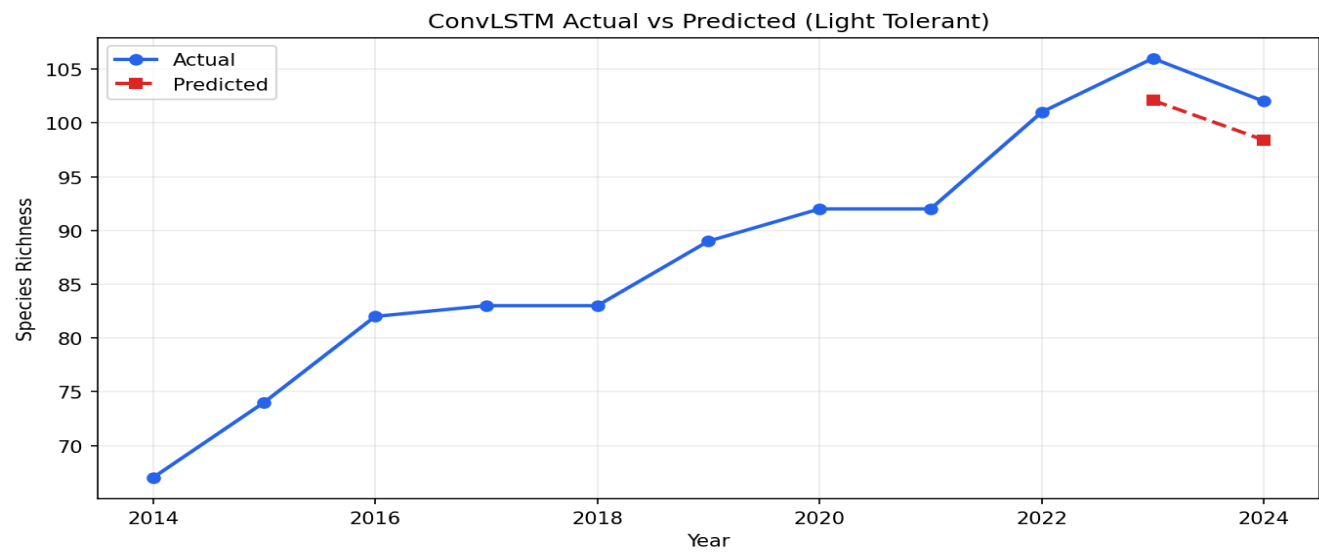
Chart: ConvLSTM Actual vs Predicted (Light Sensitive)



Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

Year		Actual	Predicted
	2014	42.0000	-
	2015	58.0000	-
	2016	55.0000	-
	2017	58.0000	-
	2018	57.0000	-
	2019	61.0000	-
	2020	72.0000	-
	2021	80.0000	-
	2022	89.0000	-
	2023	98.0000	93.8000
	2024	97.0000	93.1000

Chart: ConvLSTM Actual vs Predicted (Light Tolerant)

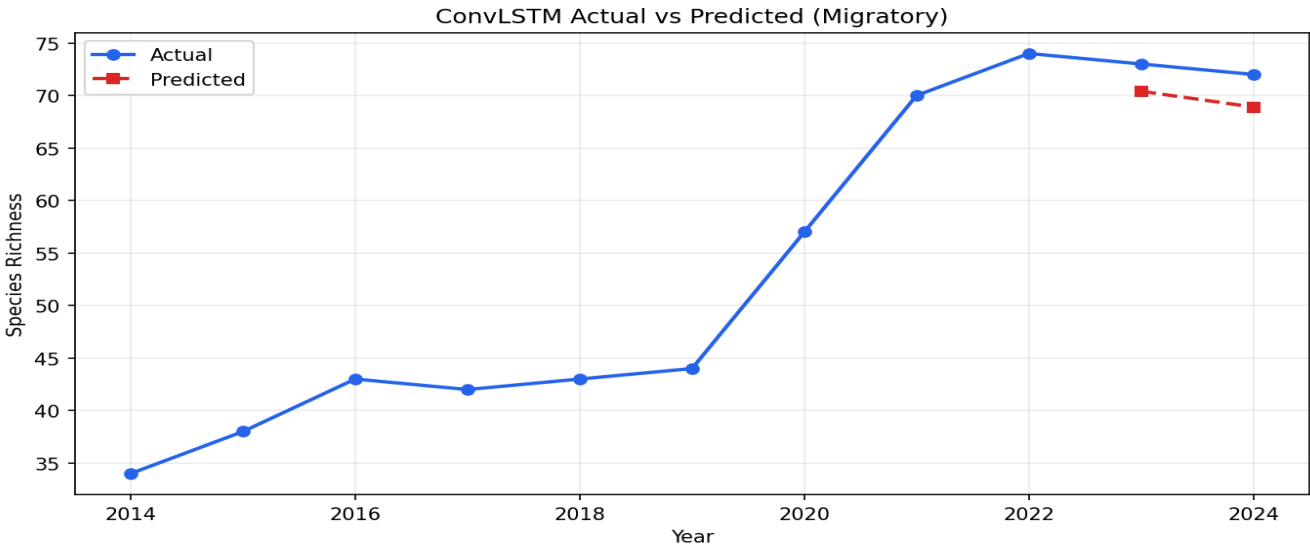


AviLight Ecological Impact and Diagnostics Report

Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

Year	Actual	Predicted
2014	67.0000	-
2015	74.0000	-
2016	82.0000	-
2017	83.0000	-
2018	83.0000	-
2019	89.0000	-
2020	92.0000	-
2021	92.0000	-
2022	101.0000	-
2023	106.0000	102.1000
2024	102.0000	98.4000

Chart: ConvLSTM Actual vs Predicted (Migratory)

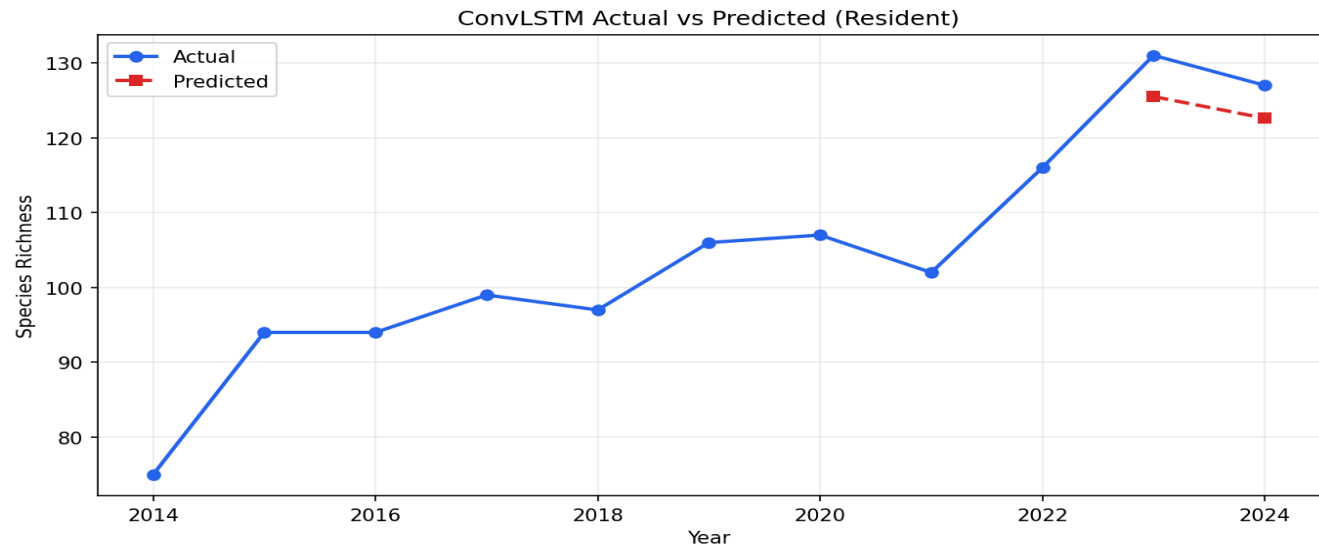


Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

Year	Actual	Predicted
2014	34.0000	-
2015	38.0000	-
2016	43.0000	-
2017	42.0000	-
2018	43.0000	-
2019	44.0000	-
2020	57.0000	-
2021	70.0000	-
2022	74.0000	-
2023	73.0000	70.4000
2024	72.0000	68.9000

AviLight Ecological Impact and Diagnostics Report

Chart: ConvLSTM Actual vs Predicted (Resident)



Numerical Values: ConvLSTM Actual vs Predicted (Resident)

Year	Actual	Predicted
2014	75.0000	-
2015	94.0000	-
2016	94.0000	-
2017	99.0000	-
2018	97.0000	-
2019	106.0000	-
2020	107.0000	-
2021	102.0000	-
2022	116.0000	-
2023	131.0000	125.5000
2024	127.0000	122.6000

Systems Recommendations: KBA/PA Audit

Site	Type	Species	Sensitive%	Light	Eff.Score
La Mesa Watershed	KBA	85	42.0	28.5	82.0
Ninoy Aquino Parks & Wildlife Center	PA	48	35.0	45.2	68.0
Las Piñas-Parañaque Critical Habitat	KBA	92	58.0	38.7	75.0
Marikina Watershed	PA	72	48.0	32.1	79.0
Laguna de Bay Wetlands	KBA	125	52.0	35.4	71.0